Kayla Mombourquette

Computer Science Graduate | Software Development, Game Development & AI

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Summary

Recent Bachelor of Applied Computer Science graduate from Acadia University, specializing in Game Development, with hands-on experience in Unity, C#, Python, Java, and C++. Experienced in developing games, building web applications, and applying machine learning to audio classification. Strong foundation in object-oriented programming, data structures, and algorithms, backed by a 4.0 GPA and Dean's List recognition. Seeking a full-time entry-level position in software development, game development, or AI.

Work Experience

06/2026 – 10/2026
Sydney, NS

Program Assistant
Immigration, Refugees and Citizenship Canada (IRCC)

- Reviewed citizenship applications to verify completeness and ensure required information and supporting documents were provided.
- Identified missing, incomplete, or incorrect information and returned applications for correction.
- Processed, digitized, and organized application records accurately while maintaining confidentiality and following established procedures.

04/2023 – 08/2024
Sydney, NS

Summer Student
Immigration, Refugees and Citizenship Canada (IRCC)

- Processed administrative documents and completed data entry tasks while maintaining accuracy and efficiency.
- Organized digital records and handled confidential information in accordance with established procedures.
- Supported daily operations by completing high-volume tasks and following document-handling guidelines.

03/2020 – 09/2022
Sydney, NS

Crew Member
Dairy Queen

- Delivered friendly customer service, processed transactions, and prepared orders in a fast-paced environment.
- Collaborated with team members to manage busy periods while maintaining service quality and food safety standards.
- Developed strong communication, multitasking, and time-management skills.

Education

09/2022 – 05/2026
Wolfville, Nova Scotia

Bachelor of Applied Computer Science – Game Development
Acadia University

- Graduated in 2026 with a 4.0 GPA.

- Dean's List, 2022–2026.
- Relevant coursework: Software Development, Artificial Intelligence, Object-Oriented Programming, Data Structures and Algorithms, Game Design and Development.

Skills

Technical Skills

- **Programming Languages:** Python, Java, C++, C#, JavaScript, HTML, CSS
- **Web Development:** React, Vue, Angular, Flask
- **Game Development:** Unity, Gameplay Programming, Game Mechanics, UI Implementation
- **Software Development:** Object-Oriented Programming, Algorithms, Data Structures, Debugging
- **AI & Machine Learning:** Audio Classification, Data Preprocessing, Model Training, Model Evaluation
- **Tools & Technologies:** Git, GitHub, Google Colab

Soft Skills

- Problem-Solving
- Analytical Thinking
- Attention to Detail
- Teamwork
- Communication
- Time Management
- Adaptability
- Organization
- Critical Thinking
- Willingness to Learn

Projects

Project Rebirth | 3D Narrative Escape Room Game

Unity, C#

- Designed and developed a 3D narrative escape room game in Unity using C#.
- Implemented mechanics such as player movement, raycast-based interaction, puzzles, and story-driven dialogue systems.
- Created immersive environments and structured progression across five levels with multiple endings.
- Integrated UI systems, audio, and game state management to support gameplay and narrative progression.

AI Fall Detection System

Python, Google Colab, Machine Learning

- Developed a machine learning system using Python and Google Colab to distinguish fall sounds from other loud noises using audio classification.
- Processed audio data and extracted MFCC features to train and evaluate classification models.
- Achieved an F1 score of 0.968 with Logistic Regression on clean audio and 0.942 with a Ridge Classifier on noisy audio.

Invasion | 3D First-Person Shooter

Unity, C#

- Designed and developed a 3D FPS game in Unity using C#.
- Implemented mechanics such as player movement, shooting, and enemy AI behaviour.
- Built game environments and integrated 3D assets to create a playable experience.

MuSu | Music Suggestion Database

Python, Flask, Spotipy, Spotify API

- Developed a web application that recommends music based on user mood and genre preferences.
- Integrated the Spotify API using Spotipy for authentication and music data retrieval.
- Implemented randomized song suggestions, album artwork, audio previews, and session-based navigation.
- Conducted user testing and incorporated feedback to improve usability and features.